and pass it over to the other services involved. By using that correlation ID, we could fetch all log messages related to a particular transaction from multiple applications.

If we follow that idea further, we'll get to *distributed tracing*, a technique for tracking requests as they flow through a distributed system, letting us localize where errors occur and troubleshoot performance issues. There are three main concepts in distributed tracing:

- A *trace* represents the activities associated with a request or a transaction, identified uniquely by a *trace ID*. It's composed of one or more spans across one or more services.
- Each step of the request processing is called a *span*, characterized by start and end timestamps and identified uniquely by the pair trace ID and *span ID*.
- *Tags* are metadata that provide additional information regarding the span context, such as the request URI, the username of the currently logged-in user, or the tenant identifier.

Let's consider an example. In Polar Bookshop, you can fetch books through the gateway (Edge Service), and the request is then forwarded to Catalog Service. The trace related to handling such a request would involve these two applications and at least three spans:

- The first span is the step performed by Edge Service to accept the initial HTTP request.
- The second span is the step performed by Edge Service to route the request to Catalog Service.
- The third span is the step performed by Catalog Service to handle the routed request.